

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Introduction to Computers

Introduction

- A computer is an electronic machine that accepts data, processes it according to a set of instructions (program) and gives out information. It is fast, accurate, diligent and tireless — but it cannot think for itself.

Characteristics of a computer

- **Speed** — performs millions of operations per second.
- **Accuracy** — results are always exact if the input and program are correct (GIGO: garbage in, garbage out).
- **Storage** — holds vast amounts of data.
- **Diligence** — it never tires or loses concentration.
- **Versatility** — it performs many different jobs with different software.
- **No intelligence of its own** — it follows instructions exactly; all 'thinking' is done by the programmer.

Data, information and processing

- Data are raw, unprocessed facts (numbers, letters, symbols); information is data that has been processed, organised and given meaning. Processing is the operation that transforms data into information. Example: 25, 18, 31 (data) → 'average = 24.7' (information).

CLASS EVALUATION

1. Define a computer.
2. Explain GIGO.
3. Distinguish between data and information with an example.
4. State three characteristics of a computer.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: History & Generations of Computers

Introduction

- The history of computing runs from counting devices to modern electronic computers. The generations of computers mark the major changes in technology, especially of the electronic components used.

Early counting devices

- Abacus — the first calculating device (beads on rods); Napier's bones — multiplication tables; Pascal's calculator (1642); Leibniz's calculator; Babbage's Difference and Analytical Engines (the first designs of a modern computer); the Jacquard loom introduced punched cards; Ada Lovelace is regarded as the first programmer.

The five generations

- **1st (1940–1956)** — vacuum tubes; huge, costly, generated much heat; machine language (ENIAC, UNIVAC).
- **2nd (1956–1963)** — transistors; smaller, cheaper, more reliable; assembly and early high-level languages (FORTRAN, COBOL).
- **3rd (1964–1971)** — integrated circuits (ICs); keyboards and monitors; operating systems appear.
- **4th (1971–present)** — microprocessors (a whole CPU on one chip); personal computers, GUIs, the internet, networks and multimedia.
- **5th (present & future)** — artificial intelligence, parallel processing, natural language and expert systems (robots, self-driving cars, machine learning).

CLASS EVALUATION

1. Name the first calculating device.
2. Which generation used vacuum tubes?
3. What is a microprocessor and which generation introduced it?
4. State two features of the fifth generation.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Computer Hardware

Introduction

- Hardware is the physical, tangible part of a computer system — everything you can see and touch. It is grouped into the system unit and its peripherals (input, output and storage devices).

Components inside the system unit

- **Motherboard** — the main circuit board connecting every component.
- **CPU (Central Processing Unit)** — the brain: control unit (directs operations), arithmetic/logic unit (calculations and comparisons) and registers; measured in gigahertz (GHz).
- **Memory** — RAM (random access memory, temporary, lost on power-off), ROM (read-only memory, permanent boot instructions), cache.
- **Storage** — hard disk drive (HDD), solid-state drive (SSD), CD/DVD, flash drive.
- **Power supply unit and fan** — electricity and cooling.
- **Ports** — USB, HDMI, VGA, audio — connections to peripherals.

Measuring computers

- Storage: bit → byte (8 bits) → kilobyte (KB) → megabyte (MB) → gigabyte (GB) → terabyte (TB) — each step × 1024. Speed: hertz, megahertz, gigahertz.

CLASS EVALUATION

1. Define hardware.
2. Name the three main parts of the CPU.
3. Differentiate between RAM and ROM.
4. Arrange in increasing order: MB, KB, TB, GB. How many bits make one byte?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Input & Output Devices

Introduction

- Input devices feed data into the computer; output devices present the processed information to the user.

Input devices

- **Keyboard** — typed text; **mouse** — pointing and clicking; **trackpad/joystick** — pointers and games.
- **Scanner** — converts pictures and documents to digital form; **webcam** — video in; **microphone** — sound in.
- **Light pen, touch screen, barcode reader, OMR (optical mark reader), OCR (optical character reader), MICR (magnetic ink character reader), biometric devices (fingerprint, face recognition)** — specialist input.

Output devices

- **Monitor (VDU)** — soft copy display, in CRT, LCD and LED types.
- **Printer** — hard copy: inkjet, laser, dot-matrix; **plotter** — large drawings and maps.
- **Speaker/headphones** — sound output; **projector** — enlarged display for a hall.
- An output device is used with its **port** and often a **driver** (software).

Choosing devices

- Consider purpose, speed, cost, quality (resolution, DPI), compatibility and durability. For example, a bank uses MICR for cheques and a laser printer for statements; a designer uses a high-resolution monitor and plotter.

CLASS EVALUATION

1. Define input and output devices.
2. Give four examples of each.
3. Which input device is used for cheques?
4. Distinguish between a monitor and a printer in terms of output type.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Storage Devices & Memory

Introduction

- Storage devices hold data and programs permanently (secondary storage), while memory holds the data the CPU is currently working on (primary storage).

Primary memory

- RAM — read/write, volatile (erased when power goes off); ROM — read-only, non-volatile, holds the boot program (BIOS); cache — very fast memory close to the CPU; registers inside the CPU.

Secondary storage

- Hard disk (HDD)** — large capacity internal storage; **SSD** — faster, no moving parts; **flash drives and memory cards** — portable; **optical discs** — CD (700 MB), DVD (4.7 GB), Blu-ray (25 GB+); **cloud storage** — data kept on remote servers, e.g. Google Drive, Dropbox.

Capacity and access

- Capacity is measured in bytes (KB, MB, GB, TB). Access is sequential (tape) or direct/random (disk). Access time is the time taken to find data; transfer rate is the speed data moves. Store important work in at least two places — the 'backup rule'.

CLASS EVALUATION

- Differentiate between primary and secondary storage.
- Why is RAM volatile?
- List two types of optical discs and their capacities.
- Mention two advantages of cloud storage.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Computer Software & Operating System

Introduction

- Software is the set of instructions that tells the hardware what to do — the intangible part of the computer. Without software, hardware is useless: hardware + software = a working computer system.

Types of software

- **System software** — runs the computer itself: the operating system (Windows, macOS, Linux, Android), utility programs (antivirus, disk cleanup, compressors), and device drivers.
- **Application software** — helps the user do jobs: word processors (Microsoft Word, Google Docs), spreadsheets (Excel), presentation software (PowerPoint), browsers (Chrome, Firefox), and packages like CorelDRAW, Photoshop. Application software may be packaged, custom, in-house or freeware/open-source.

The operating system (OS)

- The OS manages hardware and software resources and provides the user interface. Its functions include memory management, file management, process management, device management, security and providing the user interface (command-line or GUI). Popular OSes: Windows, macOS, Linux, Android, iOS.

CLASS EVALUATION

1. Define software.
2. Distinguish between system and application software with examples.
3. State four functions of an operating system.
4. Name two utility programs.